

# Jack Cutrara

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github.com/jackc625 | jackcutrara.com

## PROJECTS

### SeatWatch | Live Reservation Booking SaaS | seat.watch

2025 - Present

TypeScript, React 19, Express, PostgreSQL, Redis, BullMQ

- Architected and shipped a production SaaS across three services (React 19 SPA, Express REST API, BullMQ worker) on Railway with PostgreSQL 16 and Redis 7, totaling 56,000+ lines of TypeScript with Vitest and Playwright coverage.
- Built a dual-path booking engine delivering sub-second execution, with an automatic headless-browser fallback for graceful degradation when the primary path is unavailable.
- Engineered horizontally scalable Redis concurrency primitives: per-user circuit breakers, distributed locks, and a token-bucket rate limiter, enabling safe multi-worker autoscaling without double-booking.
- Designed a per-user execution isolation layer with Poisson-distributed, jittered poll scheduling, sustaining stable concurrent monitoring across users while avoiding thundering-herd load on upstream services.
- Delivered end-to-end Stripe billing across four tiers with transaction-level enforcement, Clerk JWT auth, AES-256-GCM credential encryption at rest, 13 React Email templates, and Sentry observability.

### Multi-Chain EVM Trading System | Automated DeFi Trading Bot | github.com/jackc625

2025 - Present

TypeScript, Node.js, Solidity, ethers v6, SQLite, React

- Built a multi-chain trading runtime in TypeScript on Node 22 spanning Ethereum, BSC, Base, and Arbitrum. Abstracted chain-specific configuration behind discriminated TypeScript unions for compile-time safety across new chains and DEXs.
- Implemented three MEV-protected transaction strategies (Flashbots bundles, bloXroute private transactions, EIP-1559 direct submission) behind a single MevStrategy interface, dispatched per-chain with strategy-specific retry and health-check protocols.
- Designed a unified exit state machine with per-token sell mutex, volatility-adjusted trailing stops, age-weighted urgency multipliers, and full state persistence to SQLite. Survives SIGKILL and resumes every position at its persisted high-water mark on restart.

### NFL Prediction System | Full-Stack ML Pipeline | github.com/jackc625

2025 - 2026

Python, FastAPI, XGBoost, DuckDB, scikit-learn, HTMX

- Designed a three-layer data pipeline (Bronze/Silver/Gold) ingesting play-by-play, market odds, and weather data. Engineered 55-70 features per game with Pydantic validation at every boundary and DuckDB + Parquet dual storage.
- Trained three specialized models: logistic regression with isotonic calibration for Win Probability, and XGBoost regressors for Against-the-Spread and Over/Under. All trained under a walk-forward temporal validation harness that hard-fails on any future-data contamination.
- Built a FastAPI + Jinja2 + HTMX dashboard with Plotly visualizations, walk-forward backtests across 2021-2024, and parallel betting simulators (flat-stake and quarter-Kelly), automated via a scheduled GitHub Actions workflow.

## PROFESSIONAL EXPERIENCE

### Software Engineer, Contract | Holloway Company, Northern Virginia

May 2026 - Present

JavaScript, TypeScript (Deno), React 18, TanStack Query, Base44 BaaS, Vitest, GitHub Actions CI

- Closed a critical cross-tenant data leak on a live multi-tenant platform by re-scoping a customer portal from client-side filtering to server-side authorization, dropping one test client's visibility from 223 jobs to its own 1, and added row-level security across all 47 data entities.
- Built the platform's test suite from 0 to ~1,400 passing checks behind a coverage-gated pre-commit hook, pinning every money and time calculation to the penny with golden-master tests before extracting it into shared modules without changing a single computed value.
- Diagnosed a production data-loss regression from an over-broad auto-archive rule and recovered 91 wrongly-archived jobs through a backup-first, human-reviewed, fully reversible process, restoring the active job list from 28 to 119.
- Designed an idempotent, geofenced time-clock as a state machine so double-taps, offline retries, and simultaneous check-ins from shared field phones reconcile to exactly one payroll entry, with correct across-midnight and missed-clock-out handling.

### Project Management Intern | Balfour Beatty, Chantilly, VA

May 2023 - Aug 2023

- Tracked deliverables and subcontractor timelines across active construction workstreams; coordinated with engineers, subcontractors, and clients to keep multi-phase milestones on schedule.
- Led stakeholder meetings and prepared status reports for project leadership, surfacing risks and blockers in real time.

## SKILLS

**Languages:** TypeScript, JavaScript, Python, Java, C++, SQL, Solidity, HTML, CSS

**Frontend:** React, Next.js, Preact, Tailwind CSS, shadcn/ui, Zustand, TanStack React Query, HTMX

**Backend & Data:** Node.js, Deno, Express, FastAPI, Spring Boot, BullMQ, PostgreSQL, Redis, SQLite, DuckDB, Prisma, Supabase, Base44 (BaaS), Pydantic, Zod

**Tools & DevOps:** Docker, Railway, AWS S3, GitHub Actions, Git, Hardhat, Playwright, Vitest, pytest, Stripe, Clerk, Sentry

**AI Dev Tools:** Claude Code, OpenAI Codex, Cursor

## EDUCATION

### Bachelor of Science, Computer Science | Western Governors University

May 2026

Previously: Virginia Tech (2020-2023). Transferred to complete CS degree.

Certifications: LPI Linux Essentials (2025)